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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/6.
  grn_tf_activity_model/grn_tf_activity_model.py
2 GRN / TF-activity model (zebrafish -> human
  orthologue mapping)
3 Loaded 22253 genes from Q1.
4
5 Loaded cached orthology map (15225 genes). Delete
  that CSV to force a re-map.
6   15225/22253 zebrafish genes mapped (68.4%).
7
8 Human gene-level vector: 13165 unique symbols.
9
10 Downloading CollecTRI regulons (human) via
   decoupler/OmniPath...
11   CollecTRI: 1185 TFs, 42990 interactions.
12
13 Inferring TF activities (ulm)...
14   Saved: tf_activity_Q1.csv
15   Significant TFs (p<0.05): 71
16   Most activated in MUT: AP1, TP53, HIF1A, JUN,
   NFKB, ETS1, SMAD3, ETS2
17   Most repressed in MUT: E2F1, E2F4, FOXJ1, E2F3,
   E2F2, HCFC1, TFDPI, ARID4B
18   Saved: tf_activity_barplot_Q1.png
19
20 Building GRN model artifact...
21   Saved: grn_model_edges.csv (3352 edges, 71 TFs
   -> 965 targets)
22   Saved: grn_model.graphml (1025 nodes, 3352 edges
   )
23
24 Sanity check vs q1_transcription_factors.csv: 5 of
   your known DE TFs also flagged by activity
   inference.
25   Overlap: AR, CEBPB, ESR1, JUN, RUNX1
26
27 Applying the SAME model to tert3m_MUT_vs_WT_results
   .csv ...
28   orthology: mapped batch 1 (2000/5443)
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29 orthology: mapped batch 2 (4000/5443)
30 orthology: mapped batch 3 (5443/5443)
31 Saved: tf_activity_dataset2.csv
32 Saved: tf_activity_barplot_dataset2.png
33
34 Process finished successfully! Outputs in: /Users/
    mehakagrawal/Desktop/Final_Dissertation/Outputs/6.
    grn_tf_activity_model
35
36 Process finished with exit code 0
37
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